

MAHEK SHAIKH

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SUMMARY

AI/ML Engineer with 15+ months of applied research and production experience across Speech & Audio AI, Trustworthy AI, Agentic AI/LLMs, Predictive Maintenance, and Computer Vision. Experienced in building self-supervised audio models, RAG pipelines, transformer-based systems, and Physics-Informed Neural Networks for Remaining Useful Life (RUL) prediction. Strong foundation in PyTorch, TensorFlow, Linux, Docker, GPU optimization, and end-to-end AI deployment with a research-driven approach to trustworthy and production-ready machine learning systems.

TECHNICAL SKILLS

Programming & Mathematics: Python, PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn, Probability, Statistics, Linear Algebra, Calculus

Speech & Audio AI: Wav2Vec2, Self-Supervised Audio Embeddings, Audio Content Classification, Environmental Sound & Music Classification, Speech Representation Learning, Bayesian Monte Carlo Dropout, Uncertainty Estimation

Predictive Maintenance: Remaining Useful Life (RUL), Battery Health Monitoring & Degradation Prediction, Health Scoring, Physics-Informed Neural Networks (PINNs), Time-Series Modeling, Anomaly Detection, Isolation Forest

LLMs & Agentic AI: LLaMA, Mistral (7B), RAG, FAISS, LangChain, Ollama, Rasa, Tool Calling, ReAct Architecture, Sentence Transformers, Prompt Engineering

Machine Learning: CNNs, Transformers, LSTM Autoencoders, Transfer Learning, Feature Engineering, Model Evaluation, Model Optimization, Experiment Design

Computer Vision: YOLOv8, Object Detection, Image Classification, Dataset Creation & Annotation

MLOps & Deployment: Docker, Git, Linux, Azure Cloud, FastAPI, Flask, Streamlit, Render Cloud, GPU Optimization, LoRA/PEFT Fine-Tuning, Model Versioning

Software Engineering: REST APIs, SSH-based Remote Development, HLD, LLD, Modular Software Design, Technical Documentation

Data & Analytics: Data Preprocessing, Web Scraping, Power BI, Tableau, Excel

PROFESSIONAL EXPERIENCE

Data Science Intern, *Clint Solutions*

06/2025 – 09/2025 | Mumbai, India

- Developed India's first government-recognized Beach Cleanup Dashboard for an NGO using **Dash, Flask, and FastAPI**, deployed on Render Cloud for real-time coastal-cleanliness monitoring.
- Built a Goa dolphin-watching dataset via web scraping and AI-based annotation, and trained a **YOLOv8** detector to 98% accuracy, tuning confidence thresholds and NMS at the dataset and inference level.
- Researched and implemented a **Physics-Informed Neural Network (PINN)** for battery degradation prediction, embedding physics-based constraints into the loss function to improve interpretability and generalization, directly applicable to RUL estimation and health scoring.
- Collaborated cross-functionally to translate research prototypes into scalable, production-ready solutions.

Machine Learning Intern, *Anubrain Technologies*

04/2025 – 05/2025 | Remote

- Contributed to applied AI research supporting PhD-level work in medical imaging and rehabilitation, including experimentation, analysis, and technical documentation.
- Developed a monkeypox skin-lesion classification model (Streamlit UI) using **CNNs and transfer learning**, achieving 94% accuracy.
- Researched stroke-rehabilitation assessment using **multi-modal deep learning**, fusing motion-capture and sensor data for clinical movement evaluation, hands-on experience with multivariate sensor fusion.

Data Science Intern, *TheOther 2 Thirds Consulting LLP*

06/2024 – 08/2024 | Mumbai, India

- Designed and deployed production-level **LLM systems using LLaMA and Mistral (7B)** integrated via the **Rasa** framework, powering conversational analytics for survey-data interpretation.
- Built **AI-driven chatbots** that automated dashboard navigation and insight explanation, increasing client acquisition for enterprise customers.
- Engineered multi-classification transformer models, raising accuracy from 30% to 95% through data curation, augmentation, and fine-tuning.
- Developed, deployed, and version-controlled projects on **Azure Cloud, Linux, and Git**, ensuring reproducibility of AI systems.

PROJECTS

Uncertainty-Aware Audio Content Classification using Wav2Vec2 | [github](#)

Ongoing Research Project: Developed an uncertainty-aware environmental sound classification framework using Wav2Vec2 self-supervised representations with Bayesian Monte Carlo Dropout, predictive entropy, variance estimation, Expected Calibration Error (ECE), and reliability analysis, improving baseline classification accuracy from 38.4% to 75%, with ongoing research focused on further enhancing uncertainty estimation and model calibration.

Agentic AI Honeypot for Scam Detection and Intelligence Extraction | [github](#) | [link](#)

Developed an LLM-powered multi-turn conversational honeypot to engage suspected scammers and extract actionable intelligence including phone numbers, bank accounts, UPI IDs, phishing links, and email addresses. Built a modular Python-Flask backend with session management, context-aware information extraction, multi-provider LLM fallback (Gemini/OpenRouter), and automated API-based reporting for real-time fraud intelligence.

Physics-Informed Neural Networks for Li-ion Battery Degradation & RUL Prediction

Ongoing Research Project: Developing a Physics-Informed Neural Network (PINN) that integrates electrochemical degradation constraints with data-driven learning to estimate battery Remaining Useful Life (RUL) and health scores. Investigating interpretable, physics-guided deep learning for predictive maintenance and battery health management.

Lung Cancer Classification Using Feature Attention Pyramid Network | [github](#)

Achieved 98.4% accuracy classifying CT-scan images using a pretrained ResNet-50 feature extractor combined with an Attention Pyramid / Feature Attention mechanism and a dense ANN classifier.

LLM & RAG Model for Financial Data Analysis

Developed a Retrieval-Augmented Generation (RAG) system using LLaMA, FAISS, and sentence embeddings to retrieve semantically relevant financial information and generate context-aware responses. Built an end-to-end pipeline for document ingestion, semantic retrieval, prompt engineering, and conversational question answering over financial datasets.

EDUCATION

Bachelor of Engineering (Hons.) in Artificial Intelligence and Data Science

University of Mumbai | 2020 – 2024 | Mumbai, India | Honours in Cyber Security | CGPA: 9.24/10

ACHIEVEMENTS

- Finalist, **Top 2% of 40,000+** participants at **AI Impact Buildathon, AI Impact Summit India 2026**; built an LLM-powered agentic AI honeypot for cyber fraud intelligence extraction.
- Represented college at University of Mumbai as an **all-rounder student** (curriculum, sports, arts & culture) among 500+ colleges.
- Qualified for multiple national-level AI hackathons hosted by IIT-BHU, NIT-Durgapur, and other IITs/NITs.
- Qualified GATE in Data Science & Artificial Intelligence, and Computer Science, in 2024, 2025, and 2026.

PUBLICATIONS & COURSES

- Authored two student-level research papers in IRJMETs (2024) on applied machine-learning systems.
- Machine Learning Specialization, Stanford University, Coursera
- Applied Machine Learning in Python, University of Michigan, Coursera
- Excel Skills for Business Specialization, Coursera
- Google Analytics for Beginners, Google